Predicting Falcon 9 First-Stage Landing Success

IBM Applied Data Science Capstone — SpaceX launch data, EDA, interactive analytics and machine learning

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https://github.com/Coder-max-gif/spacex-falcon9-capstone

Outline

- Executive Summary
- Introduction
- Methodology
- Results
  - Insights drawn from EDA
  - Launch site proximity analysis (Folium)
  - Build a dashboard with Plotly Dash
  - Predictive analysis (classification)
- Conclusion
- Appendix

Executive Summary

Methods

- Data collection: SpaceX REST API (preserved IBM API snapshot) and Wikipedia web scraping with BeautifulSoup
- Data wrangling: outcome labels mapped to binary landing class; missing values handled
- EDA: Matplotlib visualizations and 12 SQL queries (SQLite)
- Interactive analytics: Folium launch-site maps and a Plotly Dash dashboard
- Prediction: Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV (5-fold CV)

Key results

- **60/90** launches (66.7%) landed successfully, 2010–2020
- Landing success rose from **0% (2010–13)** to **90.0% (2019)**
- KSC LC-39A had the highest site success rate (**77.3%**); SSO, GEO, HEO, ES-L1 orbits 100%
- All four models scored **84.8%–88.9%** CV accuracy; **SVM** was best on CV
- Decision tree and KNN reached the highest test accuracy (**83.3%**)
- Main error: predicted landings that failed (false positives)

Introduction

Project background and context

- SpaceX advertises Falcon 9 launches at about $62 million; other providers charge upward of $165 million
- Much of the saving comes from recovering and reusing the first stage
- If we can predict whether the first stage will land, we can estimate the cost of a launch
- A competitor (Space Y) can use this to bid against SpaceX for launch contracts

Problems we want to answer

- Which factors (payload mass, orbit, launch site, flight number, booster version) influence landing success?
- How have landing outcomes changed over time?
- Where are launch sites located, and what is near them?
- Can a machine learning model predict whether the first stage will land successfully?

Section 1

Methodology

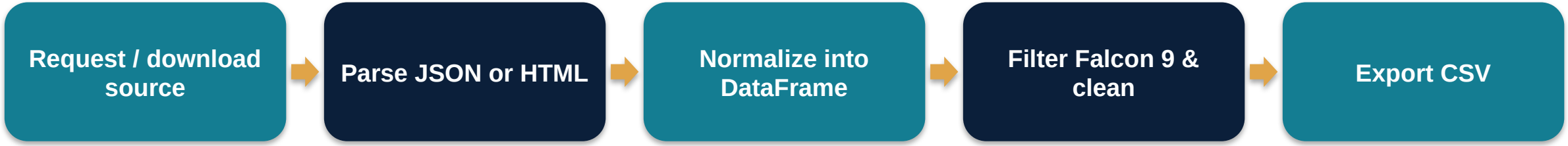
Data collection, wrangling, EDA, interactive visual analytics and predictive analysis

Methodology — Executive Summary

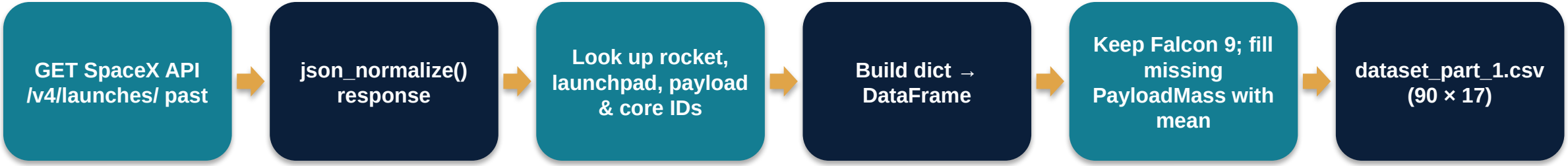
- **Data collection:** SpaceX REST API (launches, rockets, launch pads, payloads, cores) and web scraping of the Wikipedia 'List of Falcon 9 and Falcon Heavy launches' page
- **Data wrangling:** filtered to Falcon 9 launches; converted mission Outcome strings to a binary Class label (1 = landed, 0 = did not land); checked missing values
- **EDA with visualization and SQL:** scatter, bar and line charts; SQL queries on the historical launch table
- **Interactive visual analytics:** Folium maps (site markers, outcome markers, proximity lines) and a Plotly Dash dashboard
- **Predictive analysis:** scikit-learn pipelines (impute → scale → one-hot) with GridSearchCV over Logistic Regression, SVM, Decision Tree and KNN; evaluated on a stratified 20% hold-out test set

Data Collection

- Two complementary sources were used to build the launch dataset:
  - SpaceX REST API — structured JSON records for each past launch (booster, payload, orbit, site, landing outcome)
  - Wikipedia — HTML tables of Falcon 9 launch records, parsed with BeautifulSoup
- Records were combined into a pandas DataFrame, filtered to Falcon 9, cleaned and saved to CSV for EDA and modeling



Data Collection — SpaceX API



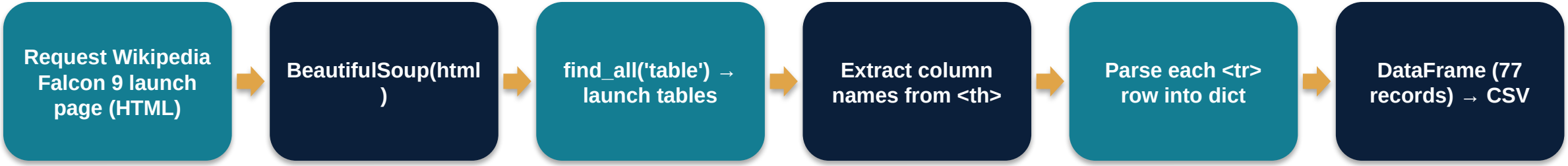

```
response = requests.get(
  'https://api.spacexdata.com/v4/launches/past')
data = pd.json_normalize(response.json())
getBoosterVersion(data); getLaunchSite(data)
getPayloadData(data);   getCoreData(data)
df = pd.DataFrame(launch_dict)
df = df[df.BoosterVersion != 'Falcon 1']
```

Notes

- Live API returned HTTP 522 at run time
- Analysis uses the preserved IBM API snapshot (90 launches, 2010-06-04 to 2020-11-05)
- Full collection code: api_collection.py

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/SpaceX_Capstone.ipynb

Data Collection — Web Scraping



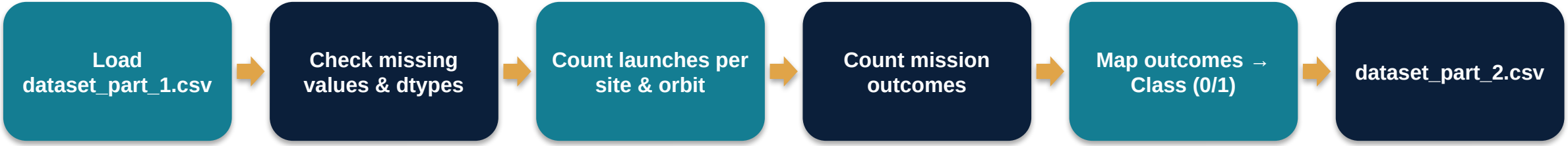

```
soup = BeautifulSoup(html, 'html.parser')
tables = soup.find_all('table', 'wikitable')
for row in table.find_all('tr'):
    cells = row.find_all('td')
    launch_dict['Booster'].append(...)
df = pd.DataFrame(launch_dict)
```

Output

- 77 Falcon 9 launch records
- Fields: flight no., date/time, booster, site, payload, mass, orbit, customer, mission and landing outcome
- Parsed from saved Wikipedia HTML

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/SpaceX_Capstone.ipynb

Data Wrangling



- Outcome combines mission result and landing type, e.g. True ASDS, False Ocean, None None
- **Bad outcomes** → Class 0: False Ocean, False RTLS, False ASDS, None ASDS, None None
- All other outcomes (True ASDS, True RTLS, True Ocean) → Class 1
- Only LandingPad had missing values (26 rows, i.e. no pad used) — kept as-is
- Result: 60 Class 1 vs 30 Class 0 (success rate 66.7%)

```
bad = {'False Ocean','False RTLS',
       'False ASDS','None ASDS',
       'None None'}
df['Class'] = (~df.Outcome
   .isin(bad)).astype(int)
df.Class.mean()  # 0.667
```

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/analyze.py (also: SpaceX_Capstone.ipynb)

EDA with Data Visualization

Chart	Variables	Purpose
Scatter	Flight Number vs Launch Site	Does experience at a site improve landing?
Scatter	Payload Mass vs Launch Site	Do heavier payloads fail more at some sites?
Bar	Success rate by Orbit	Which orbit types have the highest success rate?
Scatter	Flight Number vs Orbit	Does the orbit trend change as flights accumulate?
Scatter	Payload Mass vs Orbit	How payload affects success for each orbit
Line	Yearly success rate	Trend of landing success over time

Scatter plots show relationships between two variables coloured by outcome; bar charts compare categories; line charts show trends over time.

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/analyze.py (also: SpaceX_Capstone.ipynb)

EDA with SQL

The historical launch table (101 rows) was loaded into SQLite and queried to:

- Display unique launch site names
- Show 5 records where site begins with 'CCA'
- Total payload carried for NASA (CRS)
- Average payload for booster F9 v1.1
- Date of first successful ground-pad landing
- Boosters with drone-ship success and 4,000–6,000 kg payload
- Count of successful and failed mission outcomes
- Boosters that carried the maximum payload (subquery)
- 2015 drone-ship failures with booster and site
- Rank landing outcomes between 2010-06-04 and 2017-03-20
- Landing success counts per launch site

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Build an Interactive Map with Folium

- **Circles + markers** for each launch site (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) to show where sites are
- **Marker clusters** of launch outcomes per site, green = success (Class 1), red = failure (Class 0), to see which sites perform best
- **MousePosition** to read coordinates of nearby features
- **PolyLines + distance markers** from CCAFS SLC-40 to the nearest coastline, road and railway, using the haversine distance
- Why: to check whether site location and surroundings (coast, logistics, safety distance from cities) relate to launch success

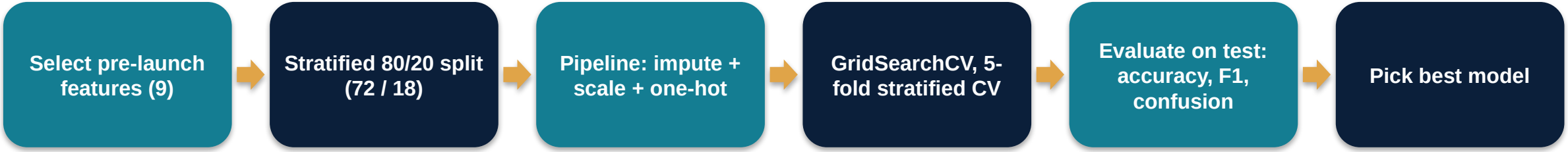
GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/SpaceX_Capstone.ipynb (also: figures)

Build a Dashboard with Plotly Dash

- **Launch site dropdown** (All sites or one site) to filter the whole dashboard
- **Pie chart:** for All sites — share of successful launches by site; for one site — success vs failure counts
- **Payload range slider** (0–10,000 kg) to select a payload window
- **Scatter plot:** payload mass vs outcome class, coloured by booster version category
- Why: lets users interactively find which sites, payload ranges and booster versions have the highest success
- Data: 56-launch IBM dashboard snapshot (spacex_launch_dash.csv)

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/dashboard.py

Predictive Analysis (Classification)



- Features: FlightNumber, PayloadMass, Flights, Block, GridFins, Reused, Legs, Orbit, LaunchSite
- Outcome-derived fields excluded to avoid leakage
- Preprocessing is fitted inside CV folds
- Extra check: train on first 72 flights, test on last 18 (time-ordered)

Model	Grid searched
Logistic Regression	C
SVM	C, kernel, gamma
Decision Tree	max_depth, min_samples_leaf
KNN	n_neighbors, weights

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/analyze.py (also: SpaceX_Capstone.ipynb)

Results

Exploratory analysis

- Success rate improves with flight number
- 2019 success rate 90.0%
- Orbit and site both matter

Interactive analytics

- All sites near coast
- KSC LC-39A leads successful launches
- Payload 2–6 t has most successes

Predictive analysis

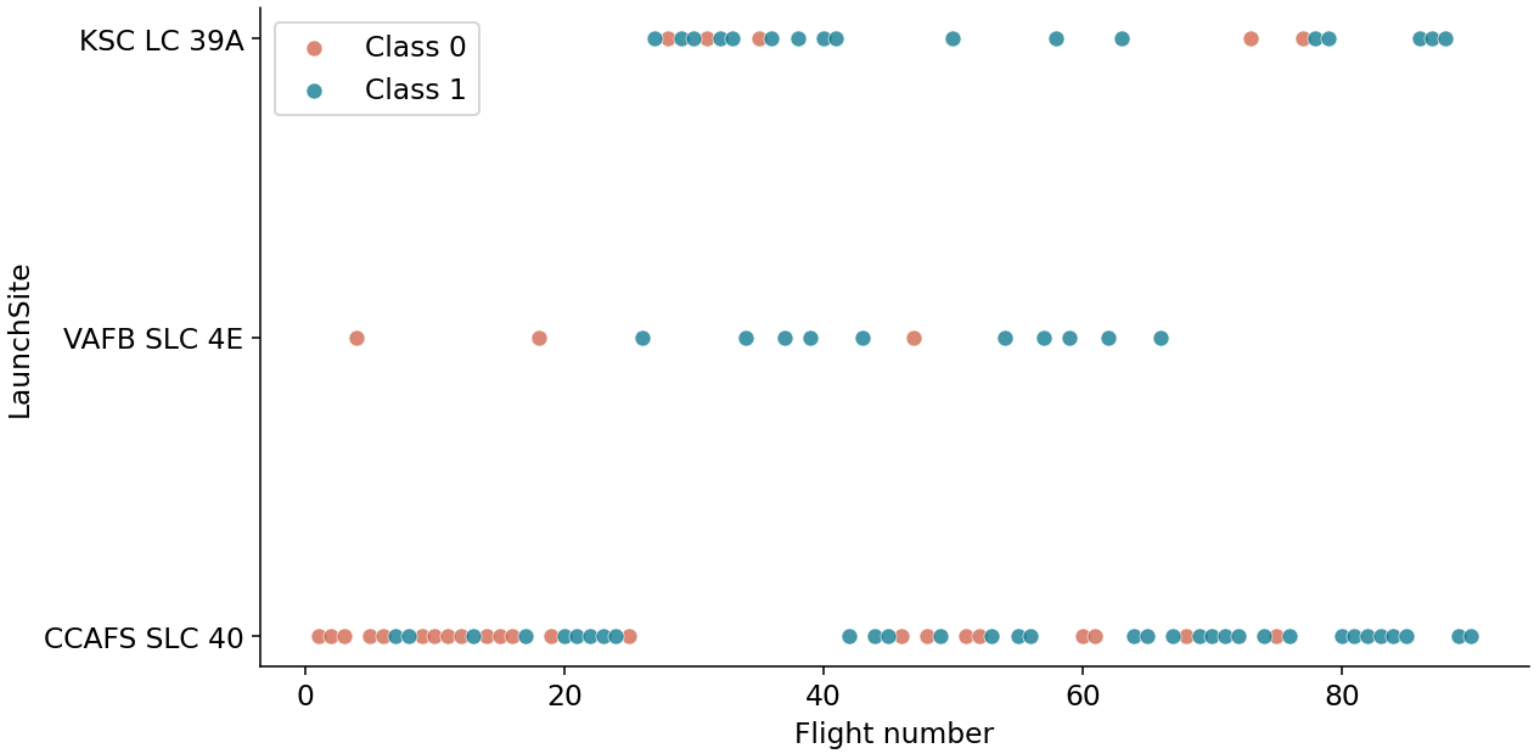
- Best CV: SVM 88.9%
- Best test: Tree/KNN 83.3%
- No false negatives on test

Section 2

Insights Drawn from EDA

Visual and SQL exploration of the Falcon 9 launch data

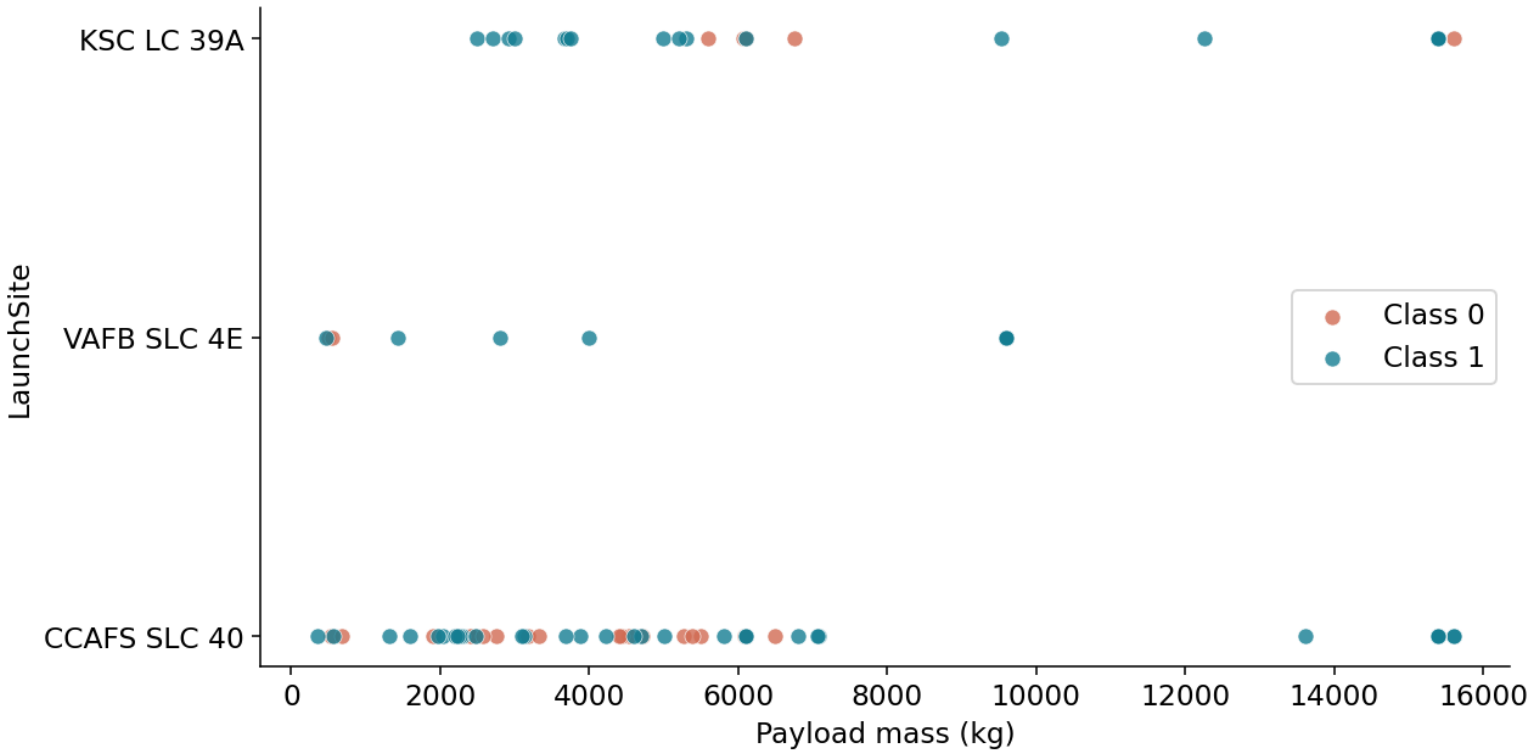
Flight Number vs. Launch Site



Insights

- Flights 1–25 (mostly CCAFS SLC 40) were largely failures
- After ~flight 40, most launches at every site succeed
- CCAFS SLC-40 has the most launches (55)
- More experience → higher landing success

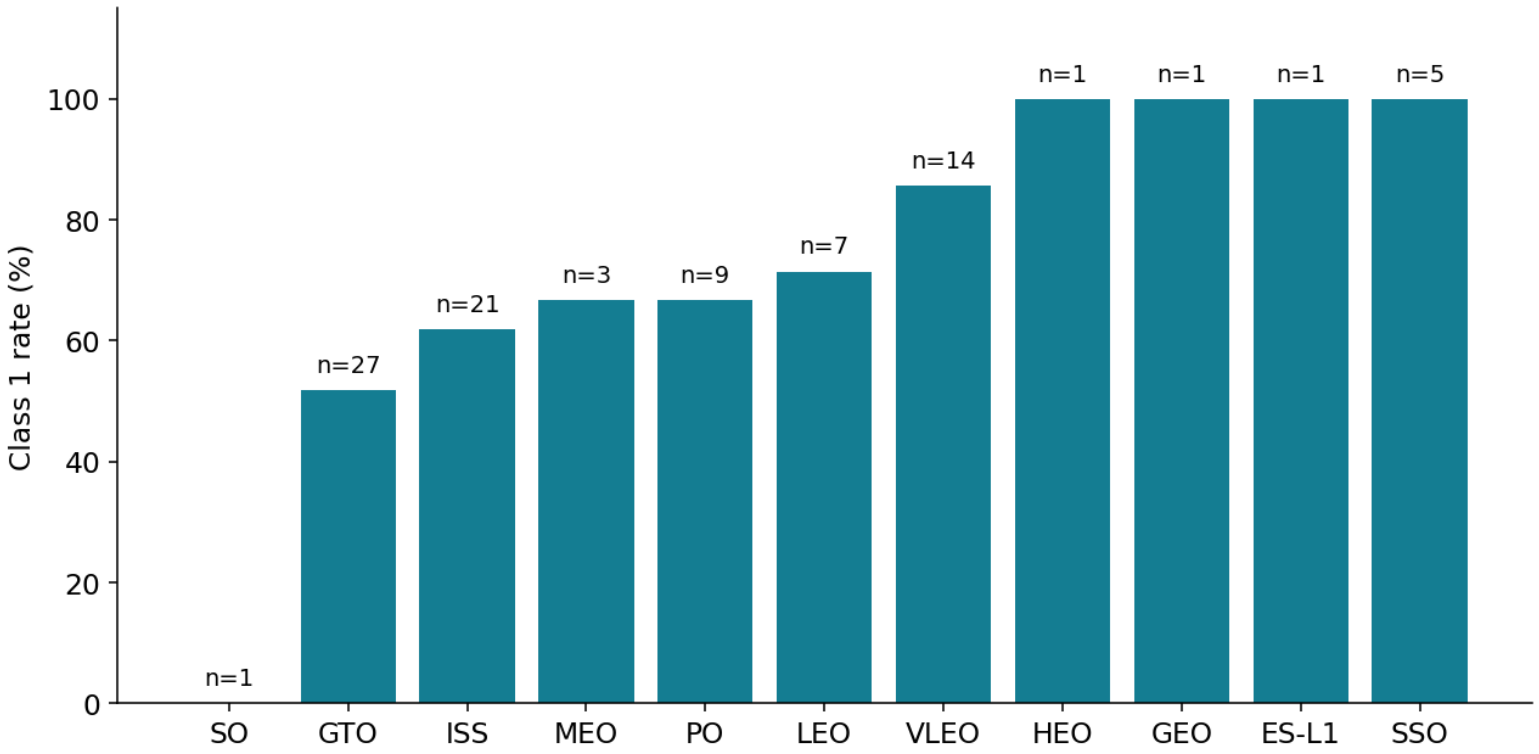
Payload vs. Launch Site



Insights

- VAFB SLC-4E launched no payloads above ~10,000 kg
- Heavy payloads (>8,000 kg) are mostly successful
- Light payloads at CCAFS show mixed outcomes
- Payload alone does not decide success

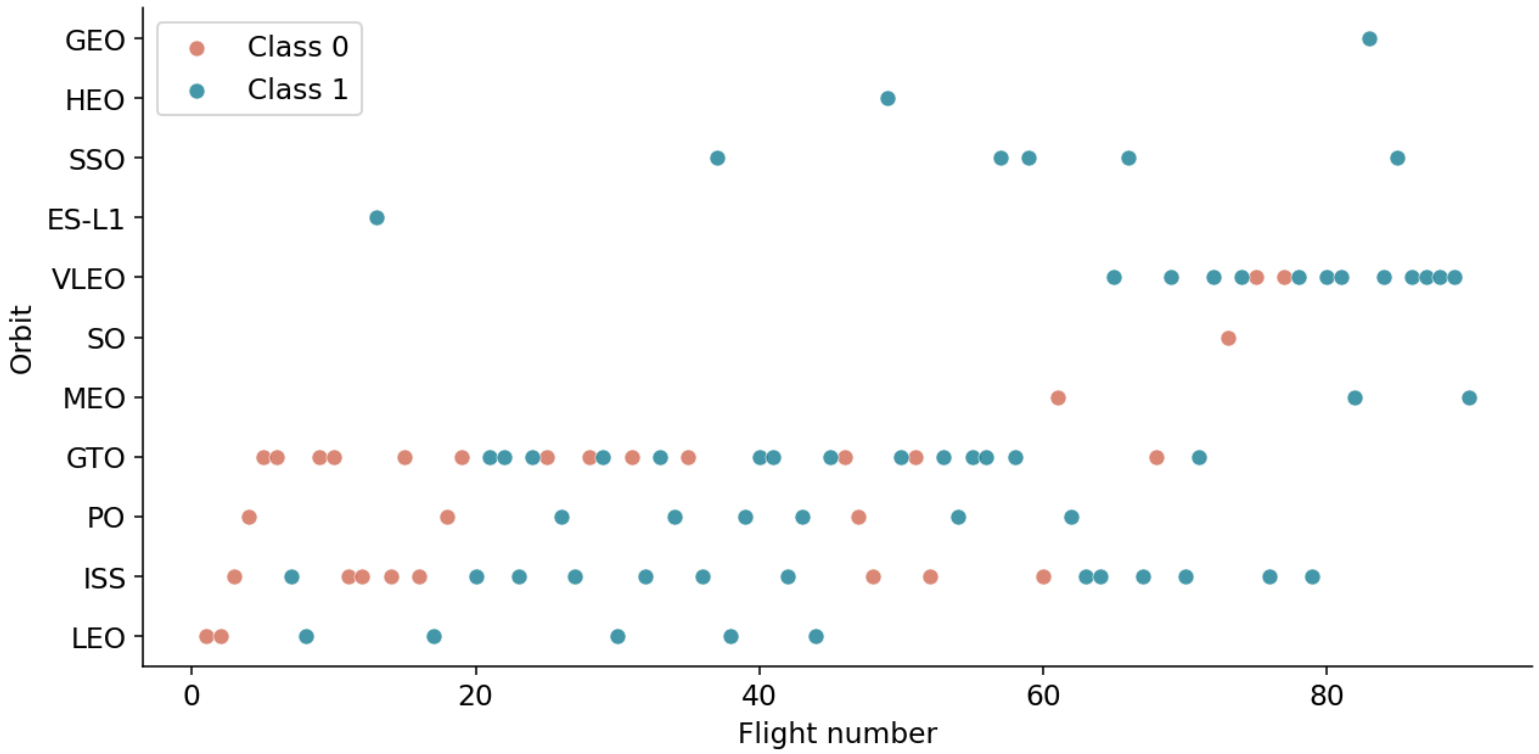
Success Rate vs. Orbit Type



Insights

- SSO (n=5), GEO, HEO and ES-L1 (n=1 each): 100%
- VLEO 85.7% on 14 launches — strongest large group
- GTO lowest of large groups at 51.9% (n=27)
- SO 0% (n=1); small groups need caution

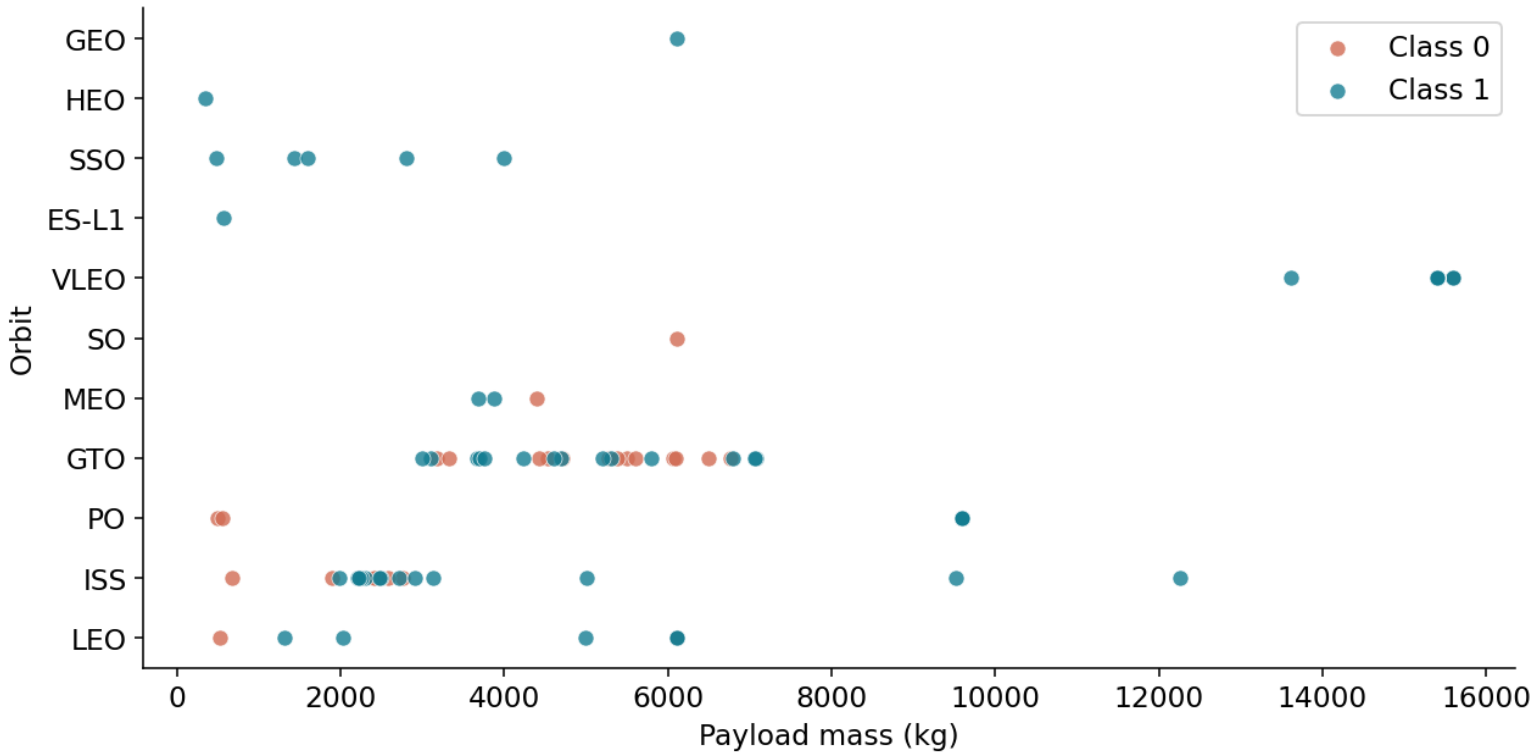
Flight Number vs. Orbit Type



Insights

- LEO success appears related to flight number
- GTO shows no clear link with flight number
- VLEO launches occur late and are mostly successful
- Later flights favour success across orbits

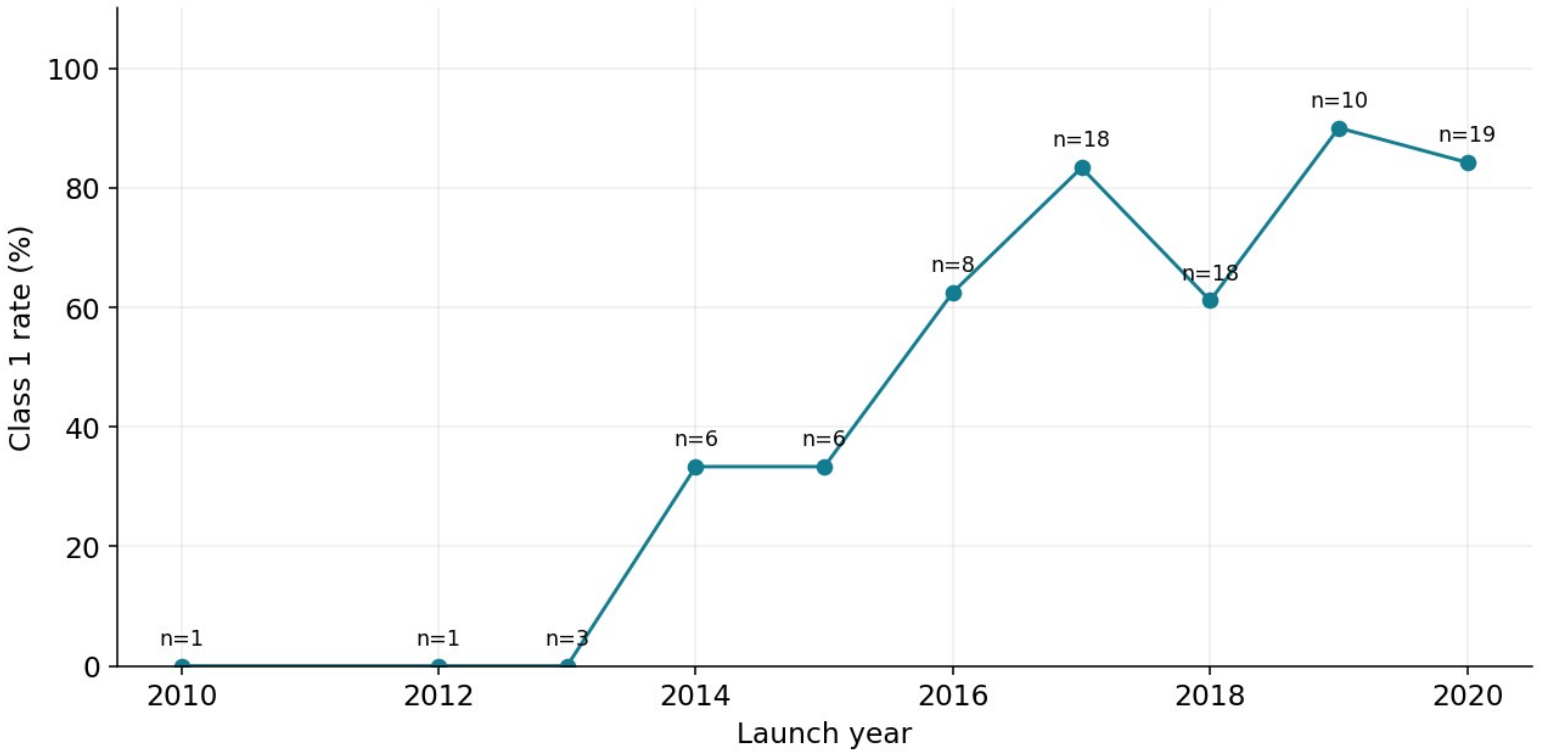
Payload vs. Orbit Type



Insights

- Heavy payloads succeed more for PO, LEO and ISS
- GTO mixes successes and failures across 3–7 t
- VLEO payloads cluster ~15 t and mostly land
- Orbit and payload interact

Launch Success Yearly Trend



Insights

- 0% success 2010–2013
- Steady rise from 2013, reaching 83.3% in 2017
- Dip to 61.1% in 2018, then 90% in 2019
- 84.2% in 2020 (n=19)

All Launch Site Names

```
SELECT DISTINCT Launch_Site FROM launches ORDER BY Launch_Site;
```

Launch_Site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Explanation

- DISTINCT returns each site once
- 4 unique launch sites: 3 in Florida, 1 in California
- CCAFS LC-40 is the older name of SLC-40

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Launch Site Names Beginning with 'CCA'

```
SELECT Date, Booster_Version, Launch_Site FROM launches
WHERE Launch_Site LIKE 'CCA%' ORDER BY Date LIMIT 5;
```

Date	Booster_Version	Launch_Site
2010-06-04	F9 v1.0 B0003	CCAFS LC-40
2010-12-08	F9 v1.0 B0004	CCAFS LC-40
2012-05-22	F9 v1.0 B0005	CCAFS LC-40
2012-10-08	F9 v1.0 B0006	CCAFS LC-40
2013-03-01	F9 v1.0 B0007	CCAFS LC-40

Explanation

- LIKE 'CCA%' matches Cape Canaveral sites
- LIMIT 5 returns the first five
- All early launches were F9 v1.0 from CCAFS LC-40

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Total Payload Mass (NASA)

```
SELECT SUM(PAYLOAD_MASS__KG_) FROM launches WHERE Customer = 'NASA (CRS)';
SELECT SUM(PAYLOAD_MASS__KG_) FROM launches WHERE Customer LIKE '%NASA%';
```

Customer filter	Total payload (kg)
NASA (CRS)	45,596
Any NASA customer	107,010

Explanation

- SUM() adds payload mass
- NASA (CRS) boosters carried 45,596 kg
- Including all NASA-related customers: 107,010 kg

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) FROM launches WHERE Booster_Version LIKE 'F9 v1.1%';
```

Booster	Average payload (kg)
F9 v1.1 (all variants)	2,534.7

Explanation

- AVG() over all F9 v1.1 boosters
- Average ≈ 2,535 kg
- Much lighter than later Block 5 loads (15,600 kg)

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

First Successful Ground Landing Date

```
SELECT MIN(Date) FROM launches WHERE Landing_Outcome = 'Success (ground pad)';
```

First successful ground-pad landing
2015-12-22

Explanation

- MIN() returns earliest date
- First ground-pad landing: 22 December 2015
- Milestone that started reliable reuse

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Successful Drone Ship Landing with Payload 4,000–6,000 kg

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM launches WHERE Landing_Outcome='Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version	Payload (kg)
F9 FT B1022	4,696
F9 FT B1026	4,600
F9 FT B1021.2	5,300
F9 FT B1031.2	5,200

Explanation

- WHERE combines landing type and payload range
- 4 boosters, all F9 FT
- Two were reused boosters (.2)

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Total Number of Successful and Failure Mission Outcomes

```
SELECT Mission_Outcome, COUNT(*) FROM launches GROUP BY Mission_Outcome;
```

Mission_Outcome	Count
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

Explanation

- GROUP BY counts each outcome
- 100 successful missions (99 + 1 unclear payload status)
- Only 1 in-flight failure
- Mission success ≠ landing success

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Boosters Carried Maximum Payload

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM launches
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM launches);
```

Booster_Version	kg	Booster_Version	kg
F9 B5 B1048.4	15,600	F9 B5 B1049.5	15,600
F9 B5 B1049.4	15,600	F9 B5 B1060.2	15,600
F9 B5 B1051.3	15,600	F9 B5 B1058.3	15,600
F9 B5 B1056.4	15,600	F9 B5 B1051.6	15,600
F9 B5 B1048.5	15,600	F9 B5 B1060.3	15,600
F9 B5 B1051.4	15,600	F9 B5 B1049.7	15,600

Explanation

- Subquery finds the max payload
- 12 launches carried 15,600 kg
- All are Block 5 boosters, many reused several times

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

2015 Launch Records

```
SELECT Date, Booster_Version, Launch_Site FROM launches
WHERE Date LIKE '2015%' AND Landing_Outcome = 'Failure (drone ship)';
```

Date	Booster_Version	Launch_Site
2015-01-10	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	F9 v1.1 B1015	CCAFS LC-40

Explanation

- Two failed drone-ship landings in 2015
- Both F9 v1.1 from CCAFS LC-40 (Jan & Apr)
- Early drone-ship attempts were difficult

GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS count FROM launches WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY count DESC;
```

Landing_Outcome	Count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

Explanation

- 'No attempt' most common (10)
- Drone ship: 5 successes, 5 failures
- Only 3 ground-pad successes in this period
- Shows the early learning curve

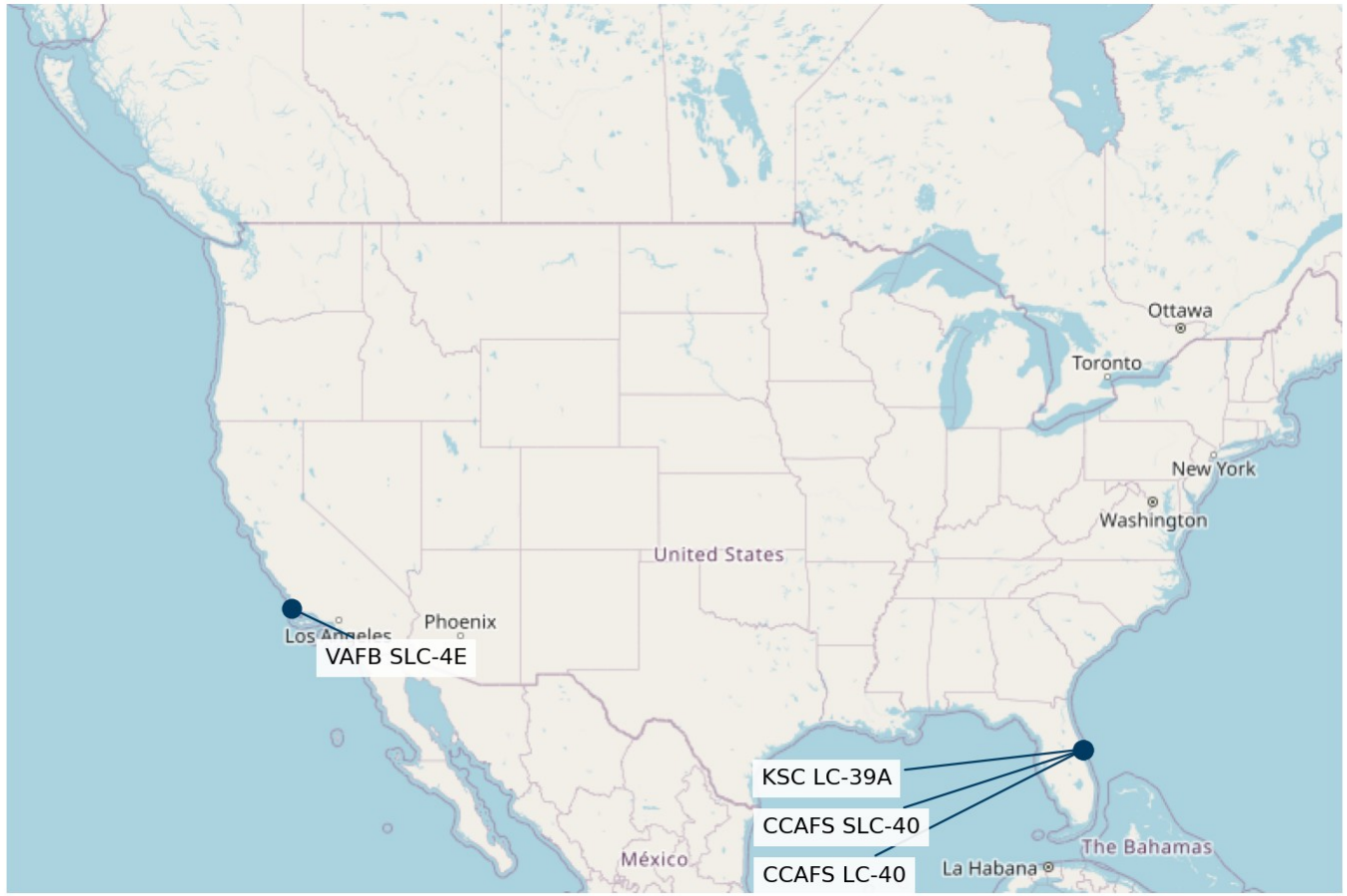
GitHub: https://github.com/Coder-max-gif/spacex-falcon9-capstone/blob/main/queries.sql (also: analyze.py)

Section 3

Launch Sites Proximity Analysis

Interactive Folium maps of launch sites, outcomes and surroundings

All Launch Sites on a Map

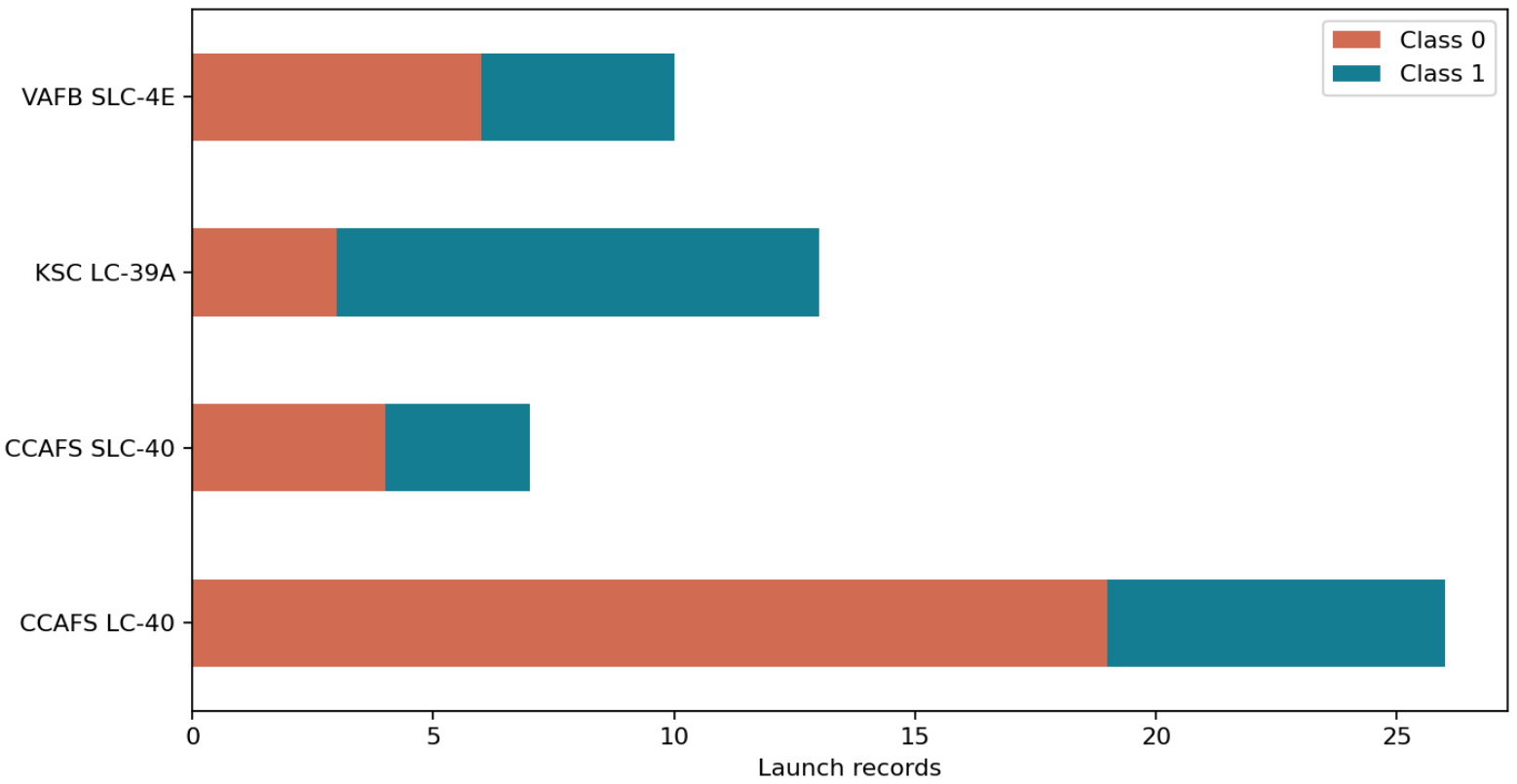


© OpenStreetMap contributors. Markers use IBM course coordinates.

Findings

- VAFB SLC-4E on the California coast
- 3 sites at Cape Canaveral, Florida
- All sites are on the coast
- Florida sites are close to the equator for extra rotational speed

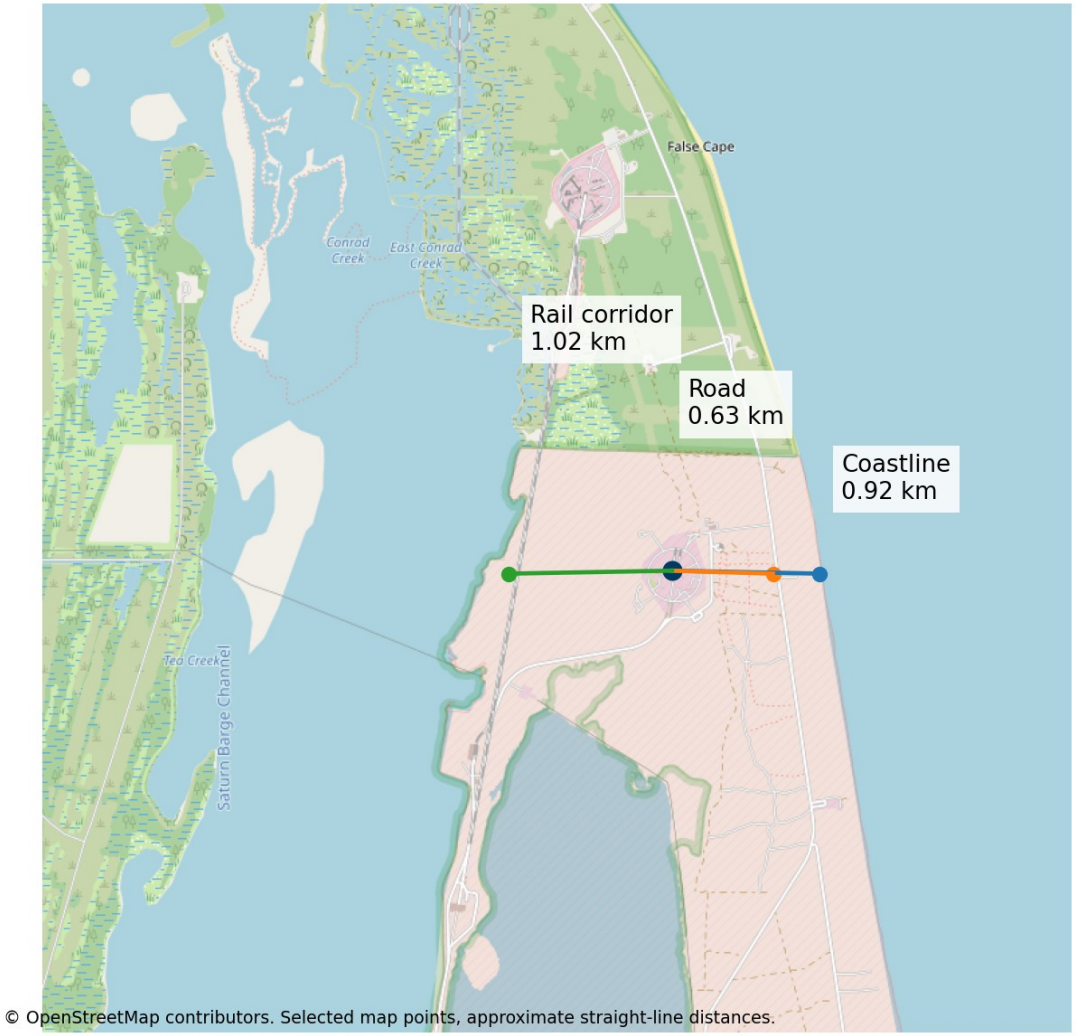
Launch Outcomes by Site (Success vs. Failure Markers)



Findings

- Folium marker clusters: green = success, red = failure; counts summarised here (teal = Class 1, orange = Class 0)
- KSC LC-39A: 10 of 13 successful (76.9%)
- CCAFS LC-40: mostly failures (7 of 26)
- Dashboard dataset: 56 launches

Launch Site Proximity to Coastline, Road and Railway



Feature (from CCAFS SLC-40)	Distance (km)
Road	0.63
Coastline	0.92
Rail corridor	1.02

Findings

- Sites are near coast for safe over-water trajectories
- Close to roads and rail for logistics
- Far from cities for safety
- Straight-line haversine distances

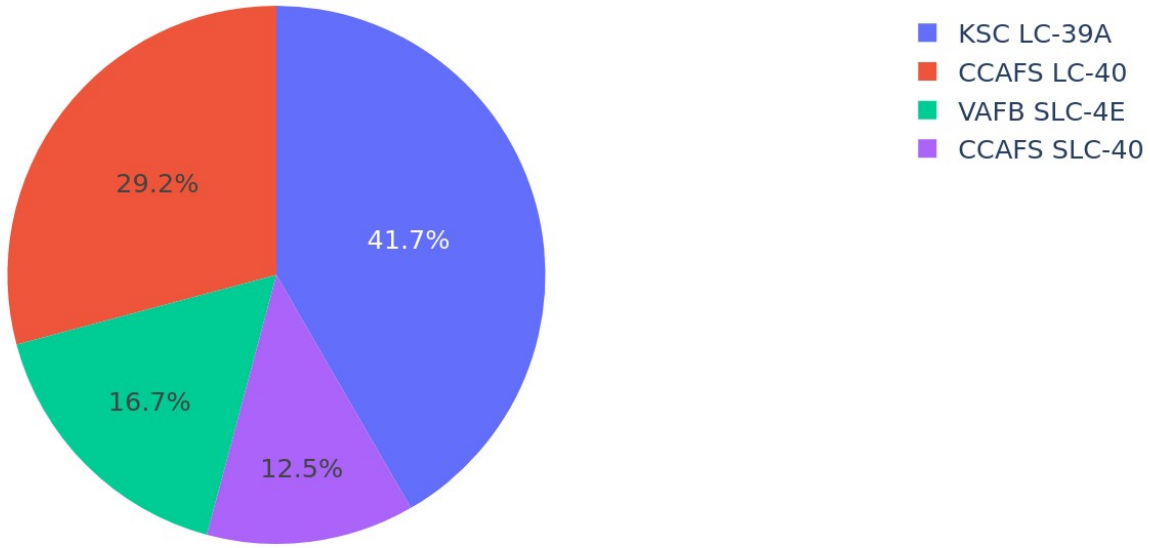
Section 4

Build a Dashboard with Plotly Dash

Interactive success and payload analysis

Launch Success Count for All Sites

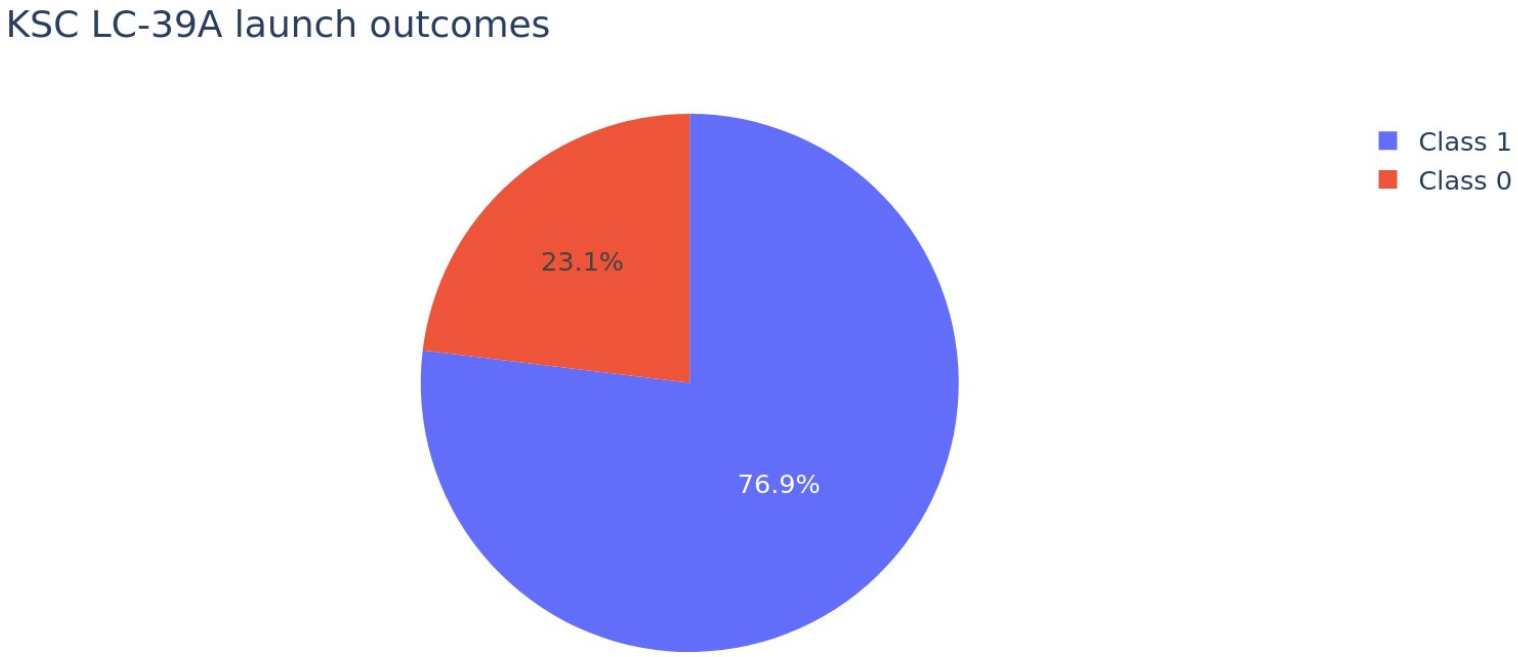
Successful launches by site (56-launch course snapshot)



Findings

- KSC LC-39A has the largest share of successful launches (41.7%)
- CCAFS LC-40 next (29.2%)
- CCAFS SLC-40 lowest (12.5%)

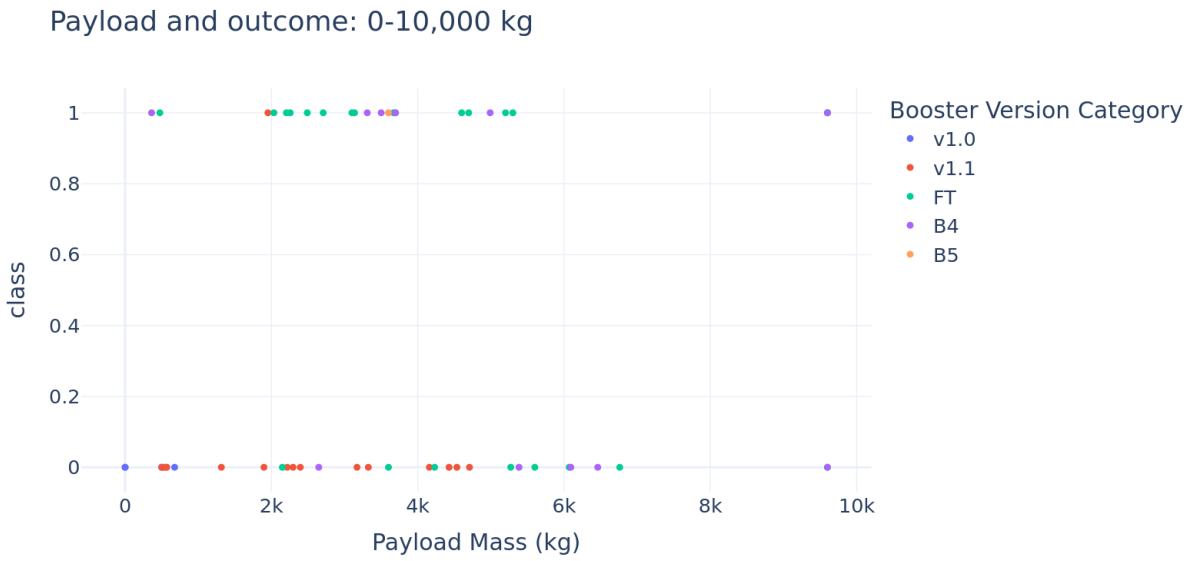
Site with Highest Launch Success Ratio — KSC LC-39A



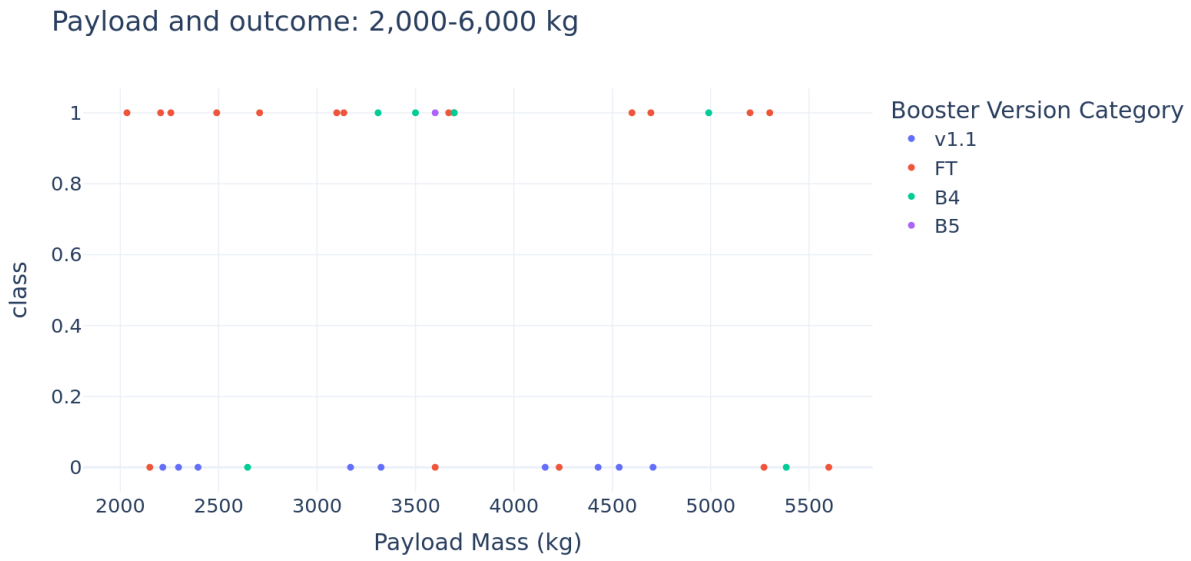
Findings

- 76.9% success (10 of 13)
- 23.1% failures (3 of 13)
- Highest success ratio of any site

Payload vs. Launch Outcome for All Sites



Full payload range 0–10,000 kg



Slider set to 2,000–6,000 kg

Findings

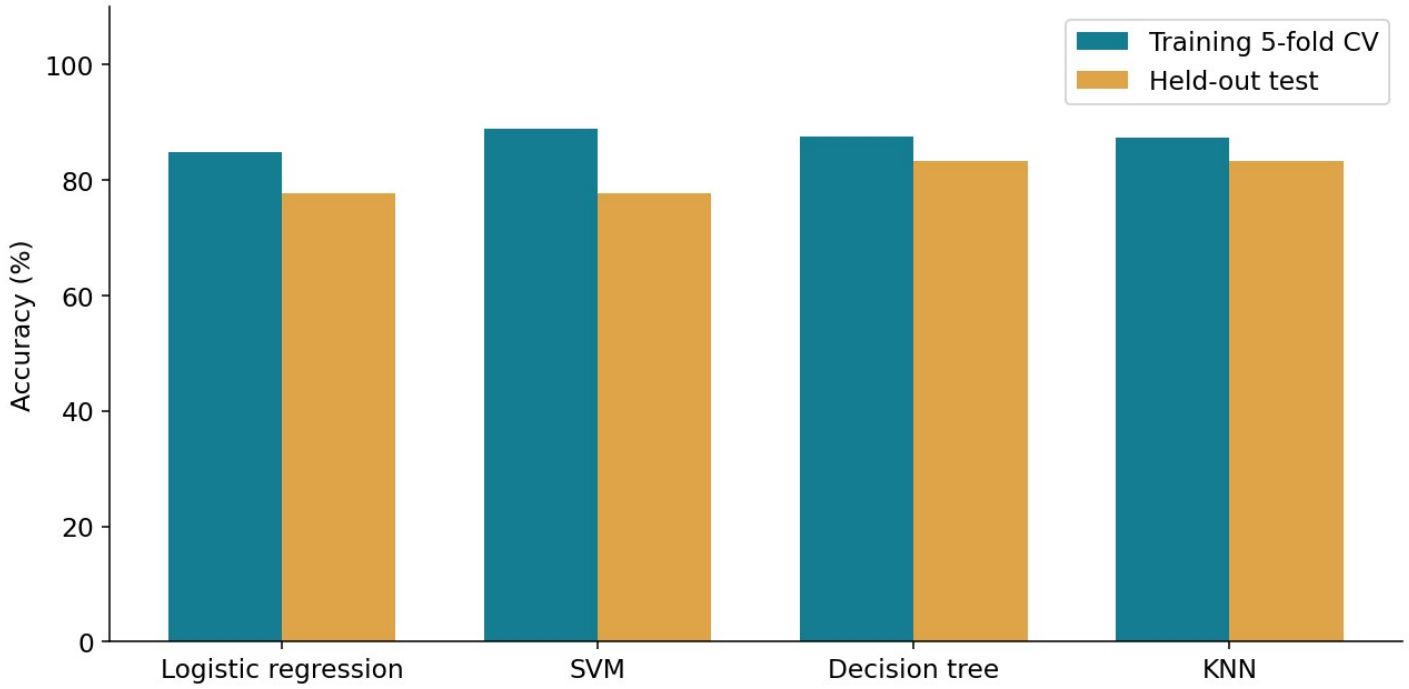
- FT boosters have the highest success rate; v1.1 boosters mostly failed
- Payloads of about 2,000–5,500 kg contain the most successful launches

Section 5

Predictive Analysis (Classification)

Model comparison and evaluation

Classification Accuracy

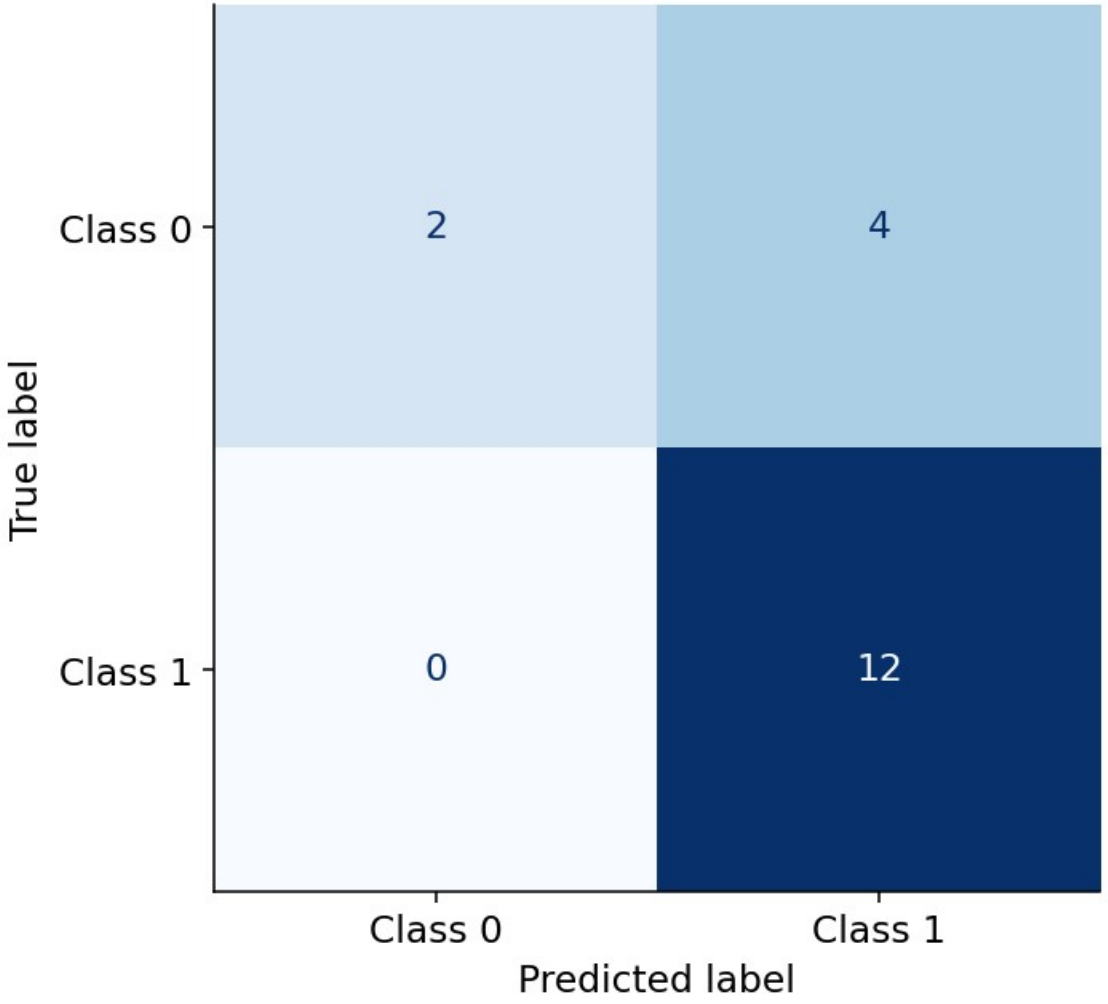


Model	CV acc.	Test acc.	F1
Logistic regression	84.8%	77.8%	0.86
SVM	88.9%	77.8%	0.86
Decision tree	87.5%	83.3%	0.88
KNN	87.4%	83.3%	0.89

Best model

- **SVM** (RBF, C=1) had the highest CV accuracy: 88.9%
- Decision tree & KNN scored highest on the test set (83.3%)
- 18-row test set: one prediction = 5.6 points

Confusion Matrix — SVM (best CV model)



Explanation

- 12 true positives: all actual landings were predicted correctly
- 0 false negatives
- 4 false positives: failures predicted as landings
- 2 true negatives
- The model is optimistic; it over-predicts landing
- Time-ordered check (last 18 flights): 88.9% vs 83.3% majority baseline

Conclusions

- Landing success increases with flight number and over time: 0% in 2010–13 → 90% in 2019
- KSC LC-39A has the highest site success rate (76.9% in dashboard data, 77.3% in modeling data)
- SSO, GEO, HEO and ES-L1 orbits had 100% success; GTO was the weakest large orbit group (51.9%)
- FT and later boosters land far more often than v1.0/v1.1
- All launch sites are near the coast, roads and rail, but away from cities
- SVM achieved the best CV accuracy (88.9%); tree-based and KNN models generalised equally well on test (83.3%)
- **Business value:** the model can estimate landing probability, and therefore launch cost, before a launch, to support Space Y bids

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What the data tells us beyond accuracy

- Model is never wrong on actual landings; it errs by being too optimistic — a costly error for a bidder
- Small test sets (18 rows) make accuracy coarse; CV and a time-ordered check give a more reliable view
- Some orbits have only 1 launch — 100% rates there are not reliable evidence

Recommendations

- Use balanced accuracy and class weights to reduce false positives
- Add booster reuse history and weather data as features
- Collect newer launches (2021+) once the API is reachable
- Report predicted probability, not just a yes/no label, for cost estimation

Appendix

- Project repository: https://github.com/Coder-max-gif/spacex-falcon9-capstone
- Files: SpaceX_Capstone.ipynb, api_collection.py, analyze.py, visuals.py, proximity.py, dashboard.py, queries.sql, results.json
- Datasets: dataset_part_1.csv (90 × 17, raw), dataset_part_2.csv (90, labelled), Spacex.csv (101 rows, SQL), spacex_launch_dash.csv (56 rows, dashboard)
- Different IBM datasets have different populations, so results are reported separately per dataset
- Map and proximity images use © OpenStreetMap contributors

